

AI61004 Statistical Foundations of AI/ML Practice Questions Module 1

Q1. Consider daily rainfall in Kharagpur as a random experiment. You are interested in the following "events" only: rainfall below 10 mm, rainfall between 10-50 mm, and rainfall above 50 mm. Define the sample space, sigma-algebra over the event space and a probability measure. Define a random variable over this probability space, and verify that it is indeed a valid random variable. Define the CDF of the random variable.

Q2. There are two coins: C1 with bias 0.6 (prob. Of HEAD), and C2 with bias 0.2. Two fair 6-faced dice D1 and D2 are rolled. If both D1 and D2 show up the same face, then C1 is tossed. But if D1 and D2 show different faces, then C2 is tossed. What is the probability of getting a HEAD from the toss?

Q3. A coin has bias 0.4. What is the expected number of times I need to toss it to get a head?

Q4. Prove that $E(X+Y) = E(X) + E(Y)$ for any two random variables X and Y.

Q5. Consider a random variable $X \sim N(a, b^2)$. Using the concept of CDF, find the PDF of $Z=(X-a)/b$

Q6. Find the mean, variance and modes of i) Beta distribution, ii) Gamma distribution using the moment-generating functions. Using the MGF, calculate their variance.

Q7. $X \sim \text{Uniform}(x,y)$. Calculate covariance between i) X and X^2 ii) X and $aX+b$

Q8. i) $X_1, X_2, \dots, X_N$ are IID random variables following Gaussian Distribution with parameters (μ, σ^2) . Define $M = (X_1+X_2+\dots+X_N)/N$. Find the mean and variance of M.

ii) Now define $Z = (X_1^2 + X_2^2 + \dots + X_N^2)/N$. Calculate its moment generating function and its mean.

Q9. X_1, X_2 are two independent Gaussian R.V.s with parameters (u_1, s_1^2) and (u_2, s_2^2) . Calculate the variance of X_1X_2 .

Q10. There are two Gaussians: $g_1(u_1, s_1^2)$ and $g_2(u_2, s_2^2)$. You toss a coin with bias p. If it is a head you sample X from g_1 , else from g_2 . Calculate the pdf, mean and variance of X.

Q11. Consider a random experiment where you are throwing two coins, with biases 0.6 and 0.4. You are interested in the “events” that i) both coins give same outcome, ii) there is exactly one ‘tail’. Define the sample space, sigma-algebra, probability space. Define a function ‘f’ by mapping each outcome to “number of heads” and analyze if ‘f’ is a valid random variable w.r.t. the probability space. [weird soln it shud be TT](#)

Q12. You are given 3 observations, which are 2D vectors of the form (X Y). The observations are [(1.7 -1.6), (2.3 -1.9), (-0.4 0.8)]. Under the assumption that they follow 2D Gaussian, estimate their mean vector and covariance matrix

Q13. You are tracking the number of Whatsapp messages you receive, along with their arrival times ($t_1, t_2, \dots$). You observe that the number N_t of messages you receive in any time-interval of length t follows a Poisson distribution with parameter $\lambda * t$. Now, consider the inter-arrival times $d_1 = t_2 - t_1$, $d_2 = t_3 - t_2, \dots$. What is the PDF of these inter-arrival times? (Hint: get the CDF first) What is the expected inter-arrival time?

Q14. Consider two PMFs: [0.25 0.3 0.15 0.3], and [0.1 a 0.1 b]. Calculate their entropies. For what values of ‘a’ and ‘b’ will the entropy of the third PMF be maximized? With respect to K-L Divergence, for what values of ‘a’ and ‘b’ will they be most “similar”?